Greenhouse challenge: grow more with a water budget

Teacher guide and worked answers

Apply plant biology to unfamiliar growth data and justify a plan using evidence and its limitations.

40 minutes. Use Lesson.html for interactive teaching or Lesson.pptx for editable slides. Slides.pdf is the static alternative. Select one pupil response route and print the shared evidence it requires. Detailed answers below are for staff.

Scheme of work

_passsl/inputs/LAUNCH KS4 - 2026-27.xlsx | LAUNCH Weekly - Spring | C38

Apply plant biology to unfamiliar data.

Directly develops C38 by comparing plant growth, light and supplied water. Core tasks use fair single-factor pairs and a budget decision. Interactions between limiting factors are explicitly optional Higher-tier stretch linked to Pearson 6.4; the core does not require inverse-square calculations.

Prior learning

Photosynthesis makes glucose that can contribute to plant biomass.

Water enters roots and can be lost by transpiration.

A fair comparison keeps other relevant conditions similar.

Success evidence

Choose a fair pair to investigate one variable.

Use numerical evidence to justify a plan against two stated conditions.

Explain a biological mechanism while distinguishing observed data from a guarantee.

Equipment

Printed shared evidence and one selected response route; pen, ruler and calculator.

Projector or offline HTML/PowerPoint. Optional word processor or spreadsheet.

Preparation

Print the shared evidence page, one chosen response route and exit page. Routes are alternatives, not a workload ladder.

Read the teacher answers before teaching. All numerical sample data in this lesson are constructed for learning.

All tables are a constructed research scenario, not results from this school. Matched plants of the same species/age were randomly assigned; temperature, nutrients and pot size were held comparable.

Dry-biomass gain is estimated from matched groups sampled at the start and after 14 days,5 plants per group. It is not repeated destructive weighing of the same living plant. No pupil drying, heating or weighing practical is required.

For optional ICT, pupils can enter the supplied table into a spreadsheet and compare two selected columns. A paper decision brief is fully equivalent.

Practical care

Paper/data activity. No heating, cutting, plant infection, soil handling or electrical practical is required.

Ready to teach alternative

All research conditions, source summaries and constructed results are supplied. Pupils can annotate a decision chart, dictate a claim or produce a paper recommendation.

Teaching sequence

Slide	Stage	Minutes	Focus
1	Opening	0-2	Can the best grower fail the brief?
2	Retrieval	2-5	Link plants with evidence
3	I do	5-9	I check what actually changed
4	I do	9-12	Read the outcome and its limits
5	I do	12-15	I use two tests before recommending
6	We do	15-18	Round 1: choose, then test the brief
7	We do	18-20	Round 2: new evidence changes the options
8	We do	20-22	Round 3: keep the recommendation honest
9	Hinge	22-25	Which claim survives both checks?
10	You do	25-26	Write a decision that can be checked
11	You do	26-36	Make the recommendation useful
12	Review	36-38	Audience challenge: what could change your mind?
13	Review	38-39	A scientific decision remains testable
14	Exit	39-40	Use the unfamiliar evidence
15	Exit	Within stage	Next useful step

Times are a suggested 40-minute route. Slides marked Within stage share the preceding stage allocation. The optional HTML timer starts only when selected.

1 Opening Can the best grower fail the brief?

Ask what “best” needs to mean. Keep the first choice provisional so revision is normal scientific reasoning, not a trick or public failure.

2 Retrieval Link plants with evidence

R1 glucose provides material for organic molecules/biomass; light supplies energy. R2 no: water has several destinations. R3 no. Use an everyday mean example if needed.

3 I do I check what actually changed

Think aloud: “I cannot isolate light by comparing A/B because their water totals also differ. C/D changes the intended light level with 0.15 L supplied in each; other stated conditions match.” Unequal plants could still create variation, so repeats/random assignment matter.

4 I do Read the outcome and its limits

Model: “B’s mean gain is 3.0 g, not 3.0 g for every plant. Matching initial/end groups estimates gain without drying the same living plant twice. This table does not directly measure instantaneous photosynthesis.”

5 I do I use two tests before recommending

Model an invented example F:2.8 g and 0.70 L. It passes growth but fails water; do not recommend it. Use diagram to show upper-left region. Keep E’s actual decision for collaborative revision.

6 We do Round 1: choose, then test the brief

B largest 3.0 g, but 0.60 L exceeds 0.50 L. A/C/D fail 2.5 g gain. No A-D plan passes both; “none yet” is a valid evidence-based decision.

7 We do Round 2: new evidence changes the options

E passes. Gain margin 2.6−2.5=0.1 g; water headroom 0.50−0.45=0.05 L. The small growth margin makes individual variation/repetition relevant. This is newly supplied constructed evidence, not a guessed interpolation.

8 We do Round 3: keep the recommendation honest

Recommend E for a repeat pilot because its mean meets both stated limits in this constructed trial. Need individual values/variation, repeated trials/more plants or independent conditions; no guarantee for each plant.

Reveal: E meets both mean-based criteria in this constructed trial. Repeat and inspect variation before a wider recommendation.

9 Hinge Which claim survives both checks?

B correct. A ignores budget; C confuses mean/individual; D confuses variables.

A. B must win because 3.0 g is the largest mean. - B exceeds the 0.50 L water-supply budget with 0.60 L.

B. E meets both mean-based criteria in the supplied trial. - 2.6 g is at least 2.5 g and 0.45 L is at most 0.50 L; scope stays with this trial.

C. A mean of 2.6 g guarantees every plant gains 2.6 g. - A mean summarises a group; individual results may differ.

D. A/B isolates light because both are plants. - Light and water supply differ between A/B; C/D is the supplied light-only pair.

10 You do Write a decision that can be checked

The independent product is a recommendation with a proposed next test. All data supplied; no online lookup required.

11 You do Make the recommendation useful

Supported S1-S4; standard M1-M4; stretch T1-T4. Label the limiting-factor interaction question optional Higher. Core pupils can succeed through fair comparisons and the two-constraint decision.

12 Review Audience challenge: what could change your mind?

A teacher, TA or quiet partner can ask. The pupil names evidence that could change the decision: repeated gains below the target, a water requirement above the budget, or large individual variation. Capture a genuine revision or justified confirmation.

13 Review A scientific decision remains testable

Model a concise recommendation using 2.6 g mean gain and 0.45 L supplied water, followed by a matched repeat and a check of individual variation. Do not infer an optimum light or water level beyond the tested plans.

14 Exit Use the unfamiliar evidence

E1: C/D is the fair pair. E2: 0.60 L exceeds the 0.50 L cap. E3: the result is a trial mean; individual variation and repeatability need checking. Match the next support to the comparison, budget or claim limit.

15 Exit Next useful step

Shared exit minute. Accept a marked decision table with an accurate spoken reason.

Answers and assessment

Retrieval

R1 photosynthesis makes glucose, which provides material for organic molecules and biomass. R2 supplied water may be retained, used or lost by several routes; it is not equal to transpiration. R3 a mean summarises a group, not every individual.

W1-W3

W1 B has greatest mean gain 3.0 g but uses 0.60 L, above 0.50 L. A/C/D fail the 2.5 g gain target; none of A-D passes both. W2 E passes with 2.6 g/0.45 L; gain margin 0.1 g and water headroom 0.05 L. W3 recommend a repeat pilot and inspect individual variation/repeats; no universal guarantee.

Hinge

A false: budget ignored. B correct with appropriate trial scope. C false: mean is not an individual guarantee. D false: A/B differs in light and water, while C/D keeps water equal.

S1 / M1

C/D: both receive 0.15 L, with low versus high light; other conditions are stated comparable. A/B changes both light and total supplied water, so cannot isolate the light effect.

S2

B:3.0 g mean gain. Its 0.60 L supplied water is 0.10 L above the 0.50 L cap, so it fails the full brief.

S3 / M2

A 1.8 g fails gain; B 3.0 g passes gain but 0.60 L fails water; C 1.0 g and D 1.1 g fail gain. E 2.6 g≥2.5 g and 0.45 L≤0.50 L, so choose E for a repeat pilot. Water headroom 0.05 L; this is a mean-based decision.

S4 / M3

Light provides energy; carbon dioxide and water are reactants used to make glucose, which contributes material to biomass. Some supplied water remains in soil/drains/is retained or used; not all is transpired. Means hide differences among plants.

M4

Repeat with randomly assigned matched groups at the same light setting, water total, nutrients, temperature and pot size; report individual values/range, not only mean. Repeated mean gain below 2.5 g, water requirement above 0.50 L or high failure variation could alter the recommendation.

T1 optional Higher

At 0.15 L, high−low gain=0.1 g. At 0.60 L, high−low=1.2 g. The response to light is larger with higher water supply, consistent with water restricting response at low supply. Dry-biomass outcomes integrate several processes; data alone do not identify a unique physiological cause. This follow-up is a separate constructed balanced comparison, not a relabelling of original A.

T2

Sum=13.0 g; mean=13/5=2.6 g. One estimate 2.2 g is below 2.5 g, so the mean target does not guarantee every plant passes.2.5 g meets an at-least 2.5 g condition.

T3

B 3.0/0.60=5 g/L supplied; E 2.6/0.45≈5.78 g/L supplied. Ratio can summarise output per supplied volume but a plan could have a good ratio while failing absolute gain or water constraints. It is not a direct physiological water-use-efficiency measurement.

T4

Use independent matched replicates, random allocation, repeated trial dates and a fixed light/water treatment; keep a named factor such as temperature/nutrients/pot size comparable and report individual variation. Do not claim one additional plant eliminates uncertainty.

Exit and assessment

E1 C/D. E2 water 0.60 L exceeds 0.50 L cap. E3 only a constructed trial mean; variation and repeatability need checking. L1 accepts a specific evidence-led revision or a justified confirmation. Secure core = fair-pair logic, both constraints and appropriately limited recommendation.

Teaching and assessment notes

40 minutes: opening 2, retrieval 3, I do 10, We do 7, hinge 3, independent 11, review 3, exit 1. Zero-minute slides share their stage time.

Choose one response route. Accept accurate speech, pointing plus explanation, annotations or writing as evidence.

A quiet review still has an audience: teacher or TA reads the pupil’s decision, asks one evidence question, and the pupil revises or confirms it.

C/D is the fair light-only pair in the original table. A/B deliberately changes both light and supplied water. The optional Higher follow-up introduces a new balanced comparison with low/high light at two fixed water supplies; do not silently change original A’s 0.30 L value.

Interactive We do has three controlled decisions: greatest mean gain, the water-cap test, and E’s follow-up results. Paper W1-W3 uses exactly the same constructed values. The diagram is a decision model, not a simulation predicting real greenhouse yields.

Dry-biomass gain is not the instantaneous rate of photosynthesis; respiration, allocation and duration contribute to the outcome. Core pupils need only connect photosynthesis with material for growth and state the evidence limit.

Access and responsive teaching

Supported

Same core scientific goal, with word bank, short prompts, read-aloud or pointing/dictation. Do not assign a route from diagnosis.

Standard

Use the supplied evidence to explain and justify an independent scientific decision.

Stretch

Optional challenge after secure core: evaluate a limitation and transfer the science. Higher-tier-only material, where present, is labelled explicitly.

Regulation

Preview the sequence. Offer solo or partner work, a pass and quiet response. Allow a brief reset and re-entry at the next numbered task; no compulsory disclosure, performance or speed competition.

Misconceptions to check

The largest growth number always wins.

A plan must meet all stated constraints, including the water-supply budget.

Check: Ask whether B’s 0.60 L exceeds the 0.50 L cap.

A mean tells us every plant grew by that amount.

It summarises a group; individual variation is not shown by the mean alone.

Check: Ask which extra data would show consistency.

Water supplied equals water transpired.

Some supplied water may remain in the soil, drain away or be used/retained; the table does not measure exact loss.

Check: Ask pupils to label the water column as supplied, not transpired.

A single successful trial guarantees future success.

Different plants, weather, pests and sampling variation can change outcomes.

Check: Request a repeat trial and a limitation before recommendation.

Word	Meaning
biomass	Mass of living material; dry biomass excludes the water removed during drying.
mean gain	Average increase across a group, not a promise for every plant.
constraint	A condition a proposed plan must meet.
replicate	A repeat experimental unit used to examine variation.
trade-off	A decision in which improving one outcome may make another harder to achieve.
limiting factor	A factor whose availability restricts a process rate.

Scientific references

Pearson Edexcel GCSE Biology 1BI0 specification

Used to check relevant Foundation curriculum content; this is teacher-authored practice, not an official assessment or complete unit. Checked 8 September 2026.

OpenStax Biology 2e: Overview of photosynthesis

Checked links between light, organic matter production and plant biomass. All greenhouse conditions and numerical outcomes are original constructed data. Checked 8 September 2026.

OpenStax Biology 2e: Transport of water and solutes in plants

Checked water supply, stomatal regulation and the distinction between transpiration and water added to soil. Checked 8 September 2026.